Spatial reconstruction: what is now supported

The paper’s descriptive direction is recovered

At P7, maturation-index values decrease with increasing graph distance from capillary ECs, proliferating ECs and pericytes under explicit reconstructed definitions. Strong correlations of distance-bin means coexist with modest cell-level correlations. This is a descriptive reconstruction, not recovery of the missing original graph, score or anatomical masks.

Shared anatomy accounts for much of the broad pattern

At P7, approximately 71% of capillary-area/maturation-index covariance lies between 200µm regions. This is a covariance decomposition, not 71% of a causal effect or variance explained. Within-region and strict matched analyses then ask whether interface-level associations remain.

Apposition relates to different components across stages

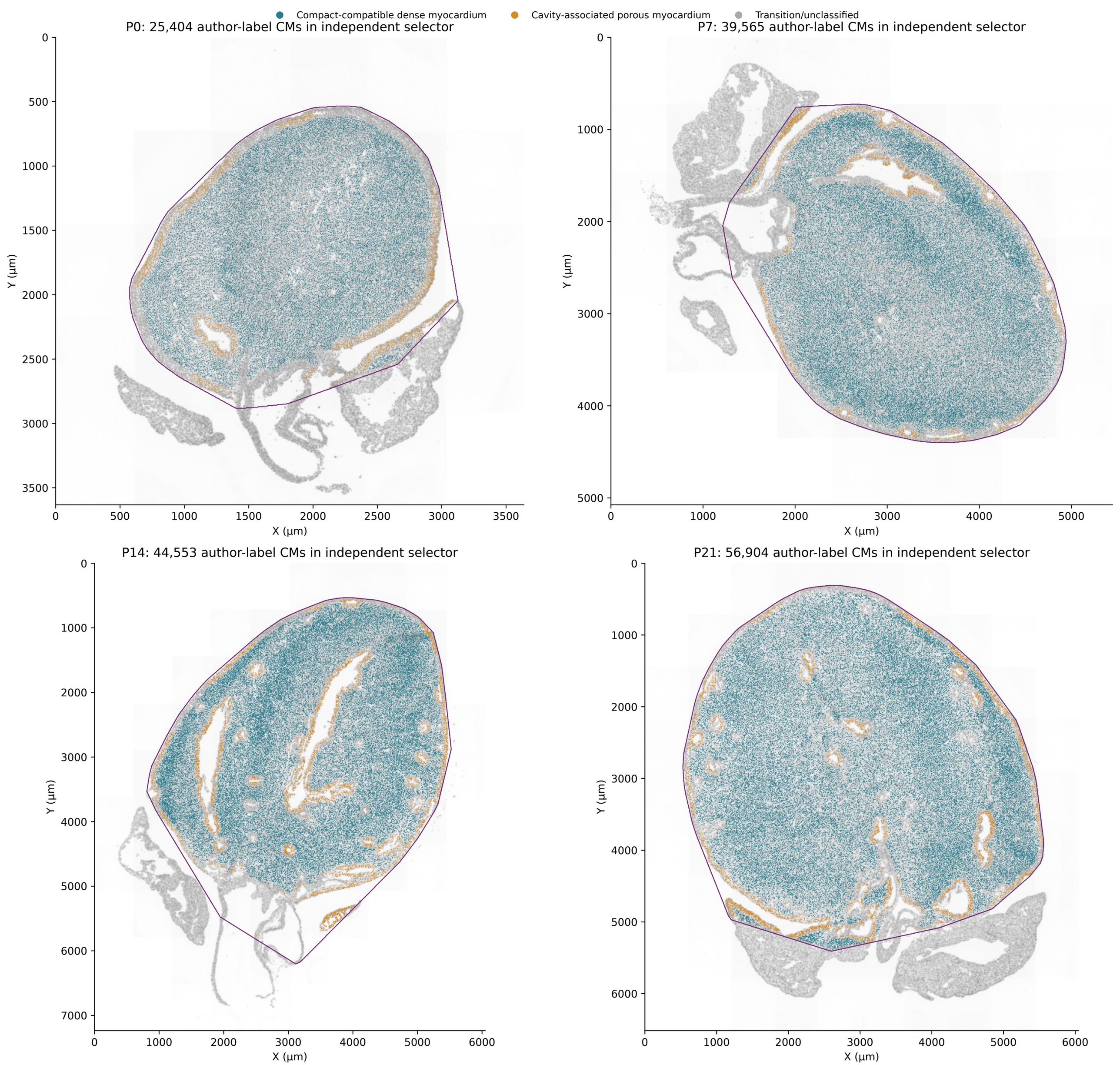
Strict matches balance local region, author CM state, capture, area, perimeter and capillary abundance. Core-RNA differences for lower cycling RNA: P7: +0.062 SD [+0.011, +0.121], 1,291 pairs. Metabolic-regulatory RNA: P14: +0.108 SD [+0.050, +0.166], 1,459 pairs; P21: +0.086 SD [+0.033, +0.143], 2,313 pairs. These small within-section contrasts motivate functional tests, not proof of a developmental switch.

RNA ordering and inferred missing days

43,249 observed ventricular RNA nuclei have a reproducible developmental ordering in pooled and disjoint replicate series. P3/P5/P10 estimates are explicitly conditional interpolation of transcript-program expectations. They do not add observed hearts, sections, perfusion measurements or biological replication.

Experiment selected

Prioritize a functional oxygen-by-contact comparison across CM developmental states, holding matrix and EC number constant. It distinguishes permissive supply from contact effects. PTPRM is a conditional molecular follow-up; the present data do not justify treating mean EC PTPRM RNA as the established driver.



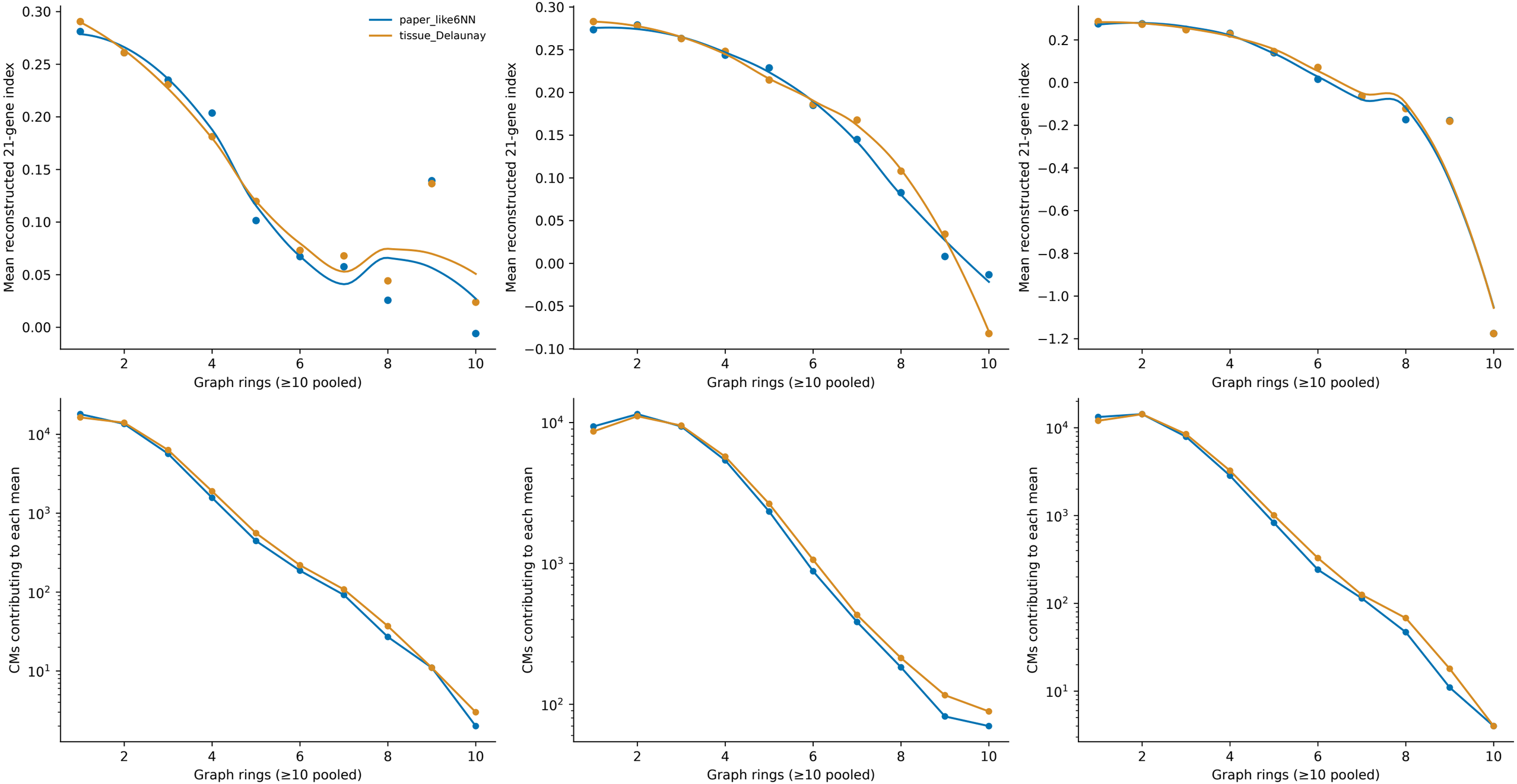
Acquired DAPI and native cell-territory geometry. Purple: independent envelope. These are operational morphology classes, not recovered author LV/RV/trabecular source labels or anatomical ground truth.

Reconstruct first: the P7 descriptive proximity pattern is positive

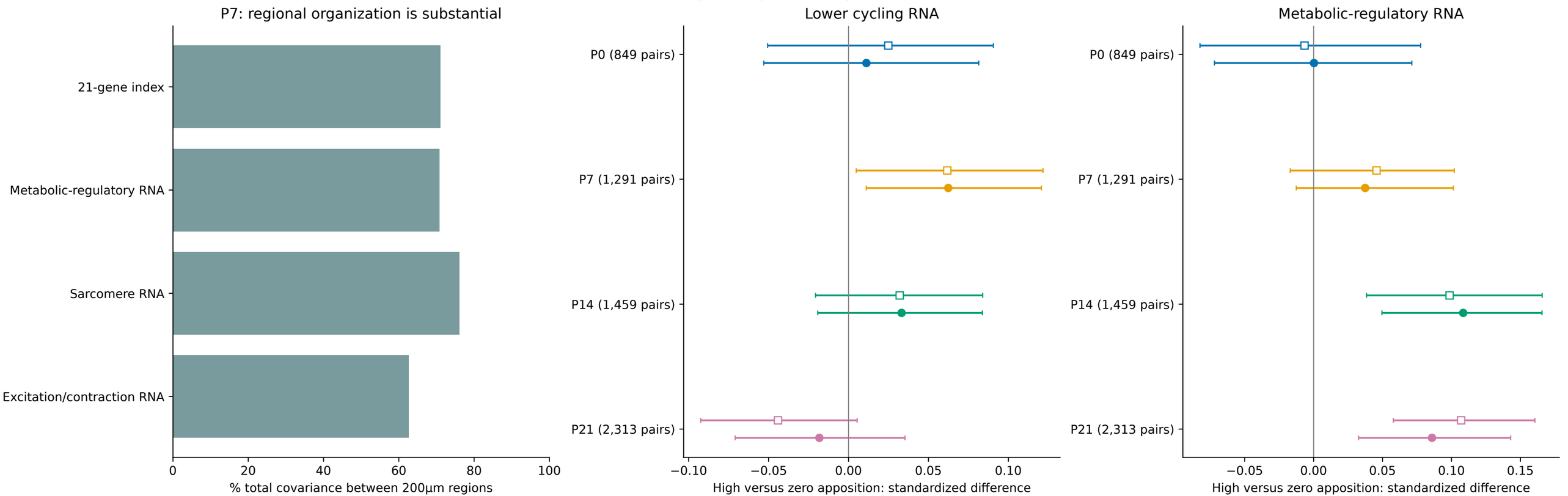
cap.EC: individual ρ=-0.074
Distance-bin mean r=-0.883

p.EC: individual ρ=-0.062
Distance-bin mean r=-0.956

Peri: individual ρ=-0.064
Distance-bin mean r=-0.813



Native measured RNA with pooled per-gene standardization; R stats::loess reproduces the released plotting rule. Exact author score/mask/neighbor inputs are absent. Bin correlations are not independent-animal evidence.



Squares: native RNA; circles: core RNA; 95% intervals from 400 µm blocks. Same 200µm region/CM state, ≤100µm apart; strict capture/shape and capillary-area calipers.
Covariance partition is not causal attribution.

Geometry sensitivity: same CMs, different exposure definitions

Native surface fraction
Refined contact count
Native ≤1µm contact count

Adjusted apposition–metabolic coefficient (SD)

0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
−0.025

P0
P7
P14
P21

Early shared cycling survives the nuclear/QC source check

Identity sensitivity: strict capillary source attenuates effects

Public cap.EC
Public cap.EC + High.Myo
Strict nuclear-supported capillary

Adjusted apposition–metabolic coefficient (SD)

0.06
0.04
0.02
0.00
−0.02

P0
P7
P14
P21

Late low cycling limits interpretation of withdrawal effects

QC-EC nuclear cycling within 50µm
QC-EC nuclear cycling in 30–50µm shell

EC cycling–CM cycling coefficient (SD)

0.06
0.04
0.02
0.00
−0.02

P0
P7
P14
P21

CM nuclear profiles
QC EC nuclear profiles

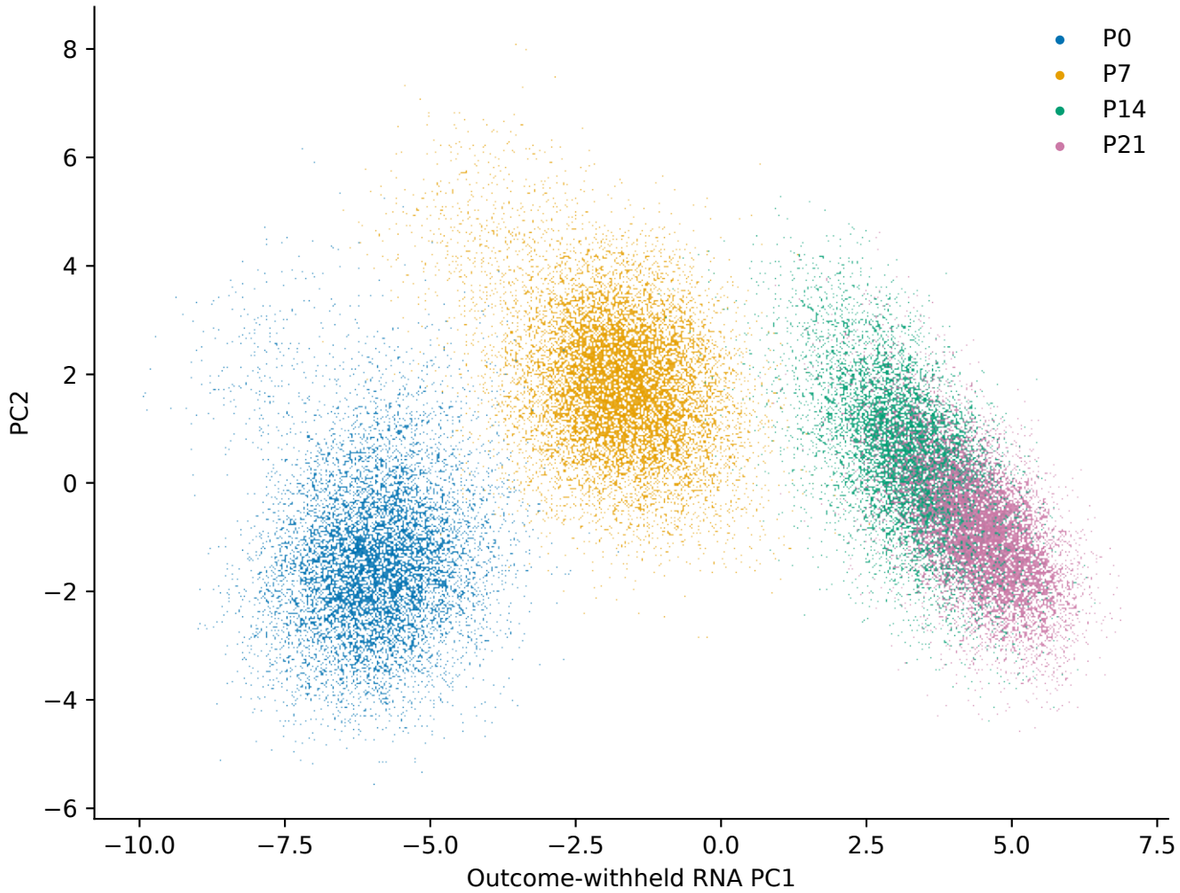
% with ≥2 measured cycling genes

20
15
10
5
0

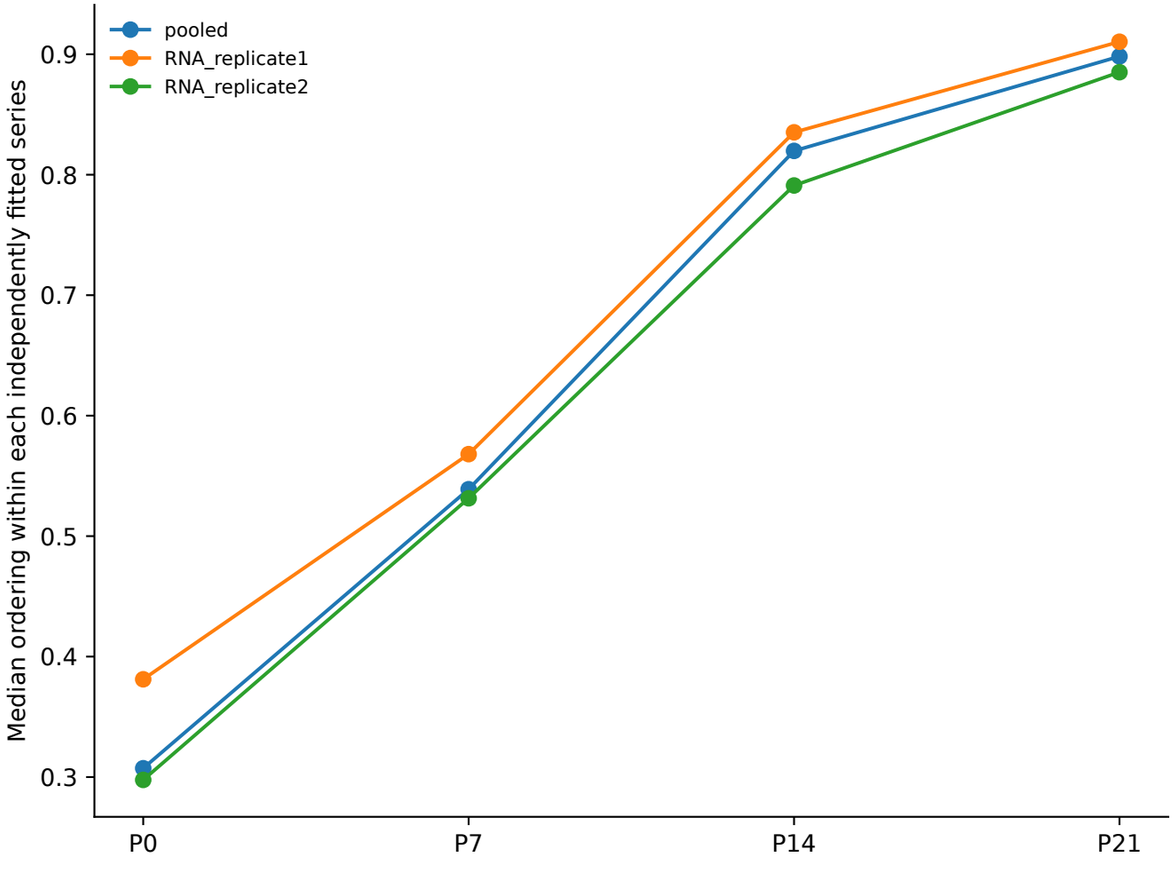
P0
P7
P14
P21

All intervals are within-section 400µm block intervals, not animal replication. Native/refined metrics have different units before standardization. Nuclear-marker coherence is not proof of completed cell division; 30–50µm shell is a centroid-distance sensitivity.

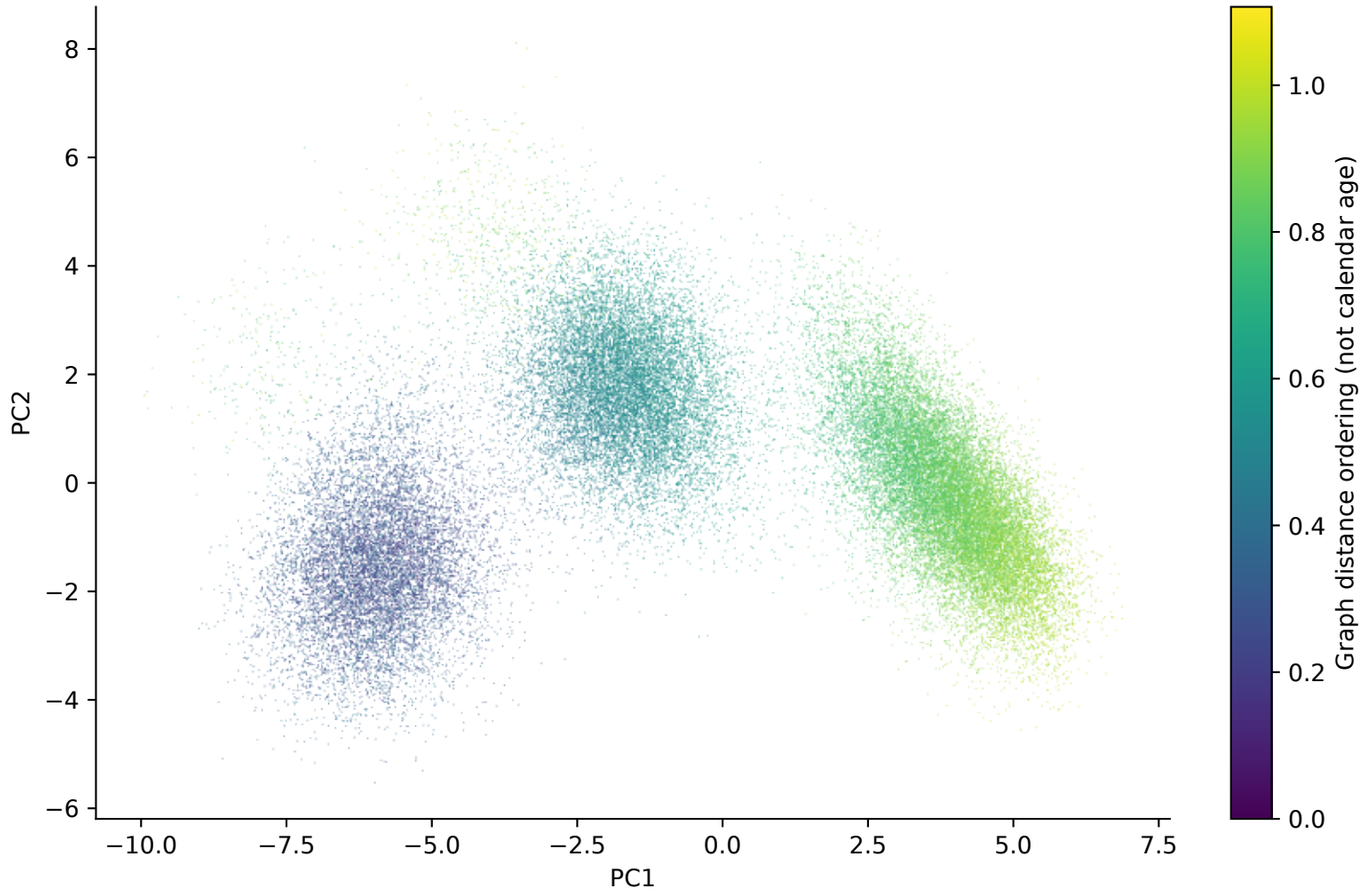
43,249 observed ventricular nuclei



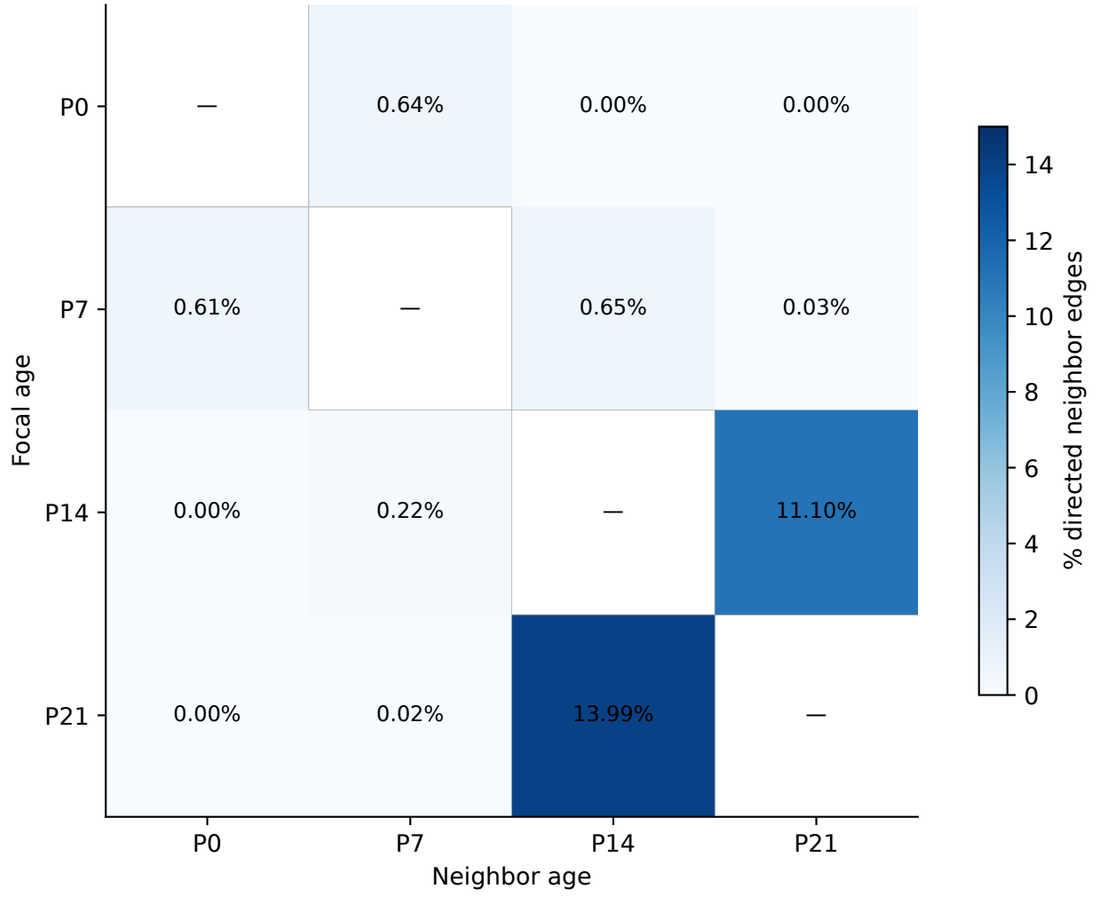
Observed order reproduces in disjoint RNA replicate series



P0-rooted ordering; no intermediate cells generated

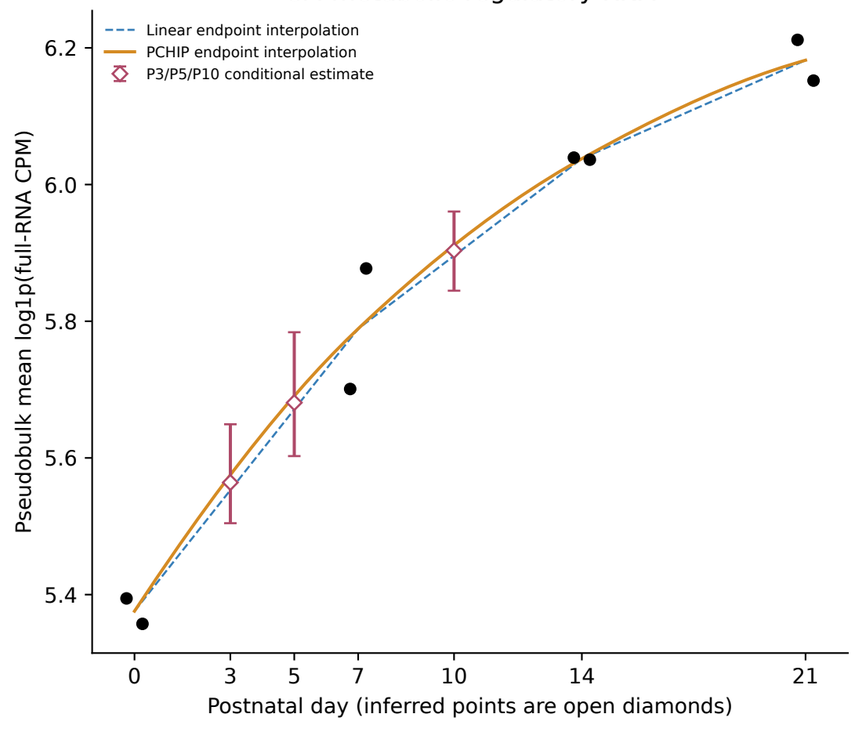


Early age gaps are sparsely bridged

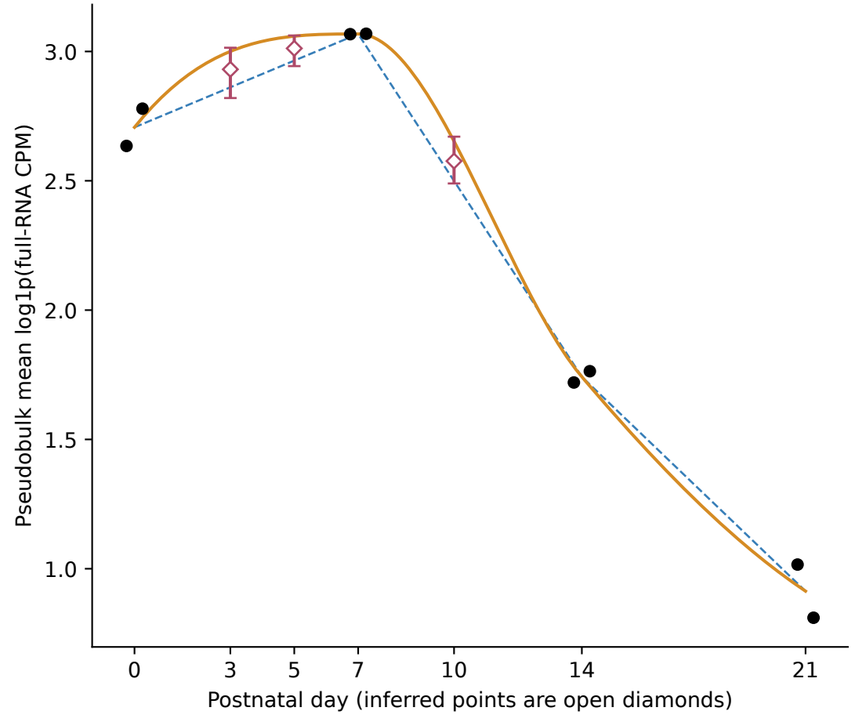


Features exclude all validation modules and cycling genes; PCA training is library-balanced. Pooled ordering removes within-age library mean offsets without forcing age integration. Root/age association is not a clock, lineage proof or dense observation of missing intervals; depth remains associated with ordering.

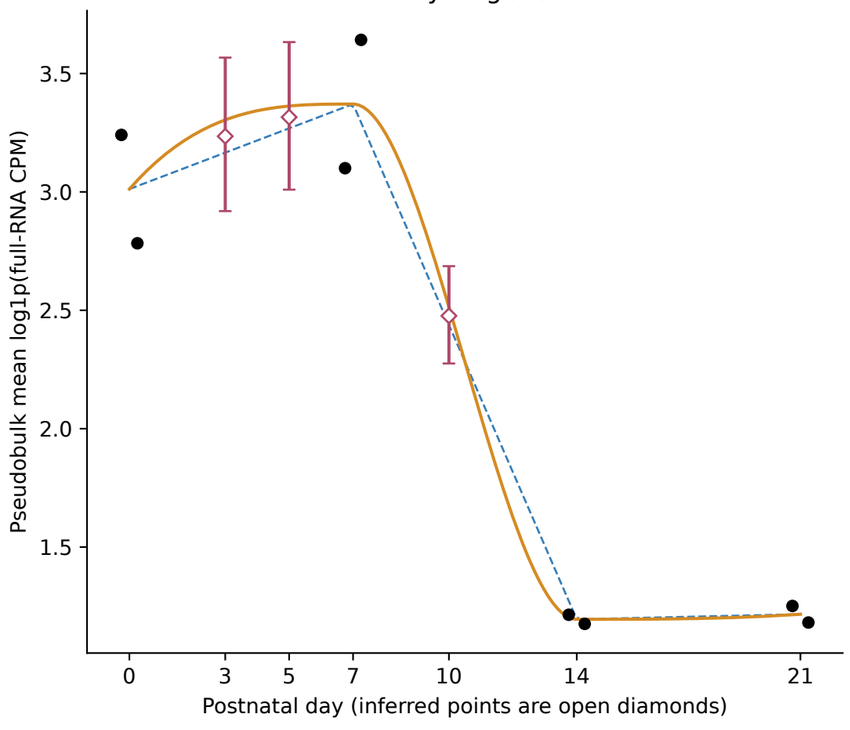
CM metabolic-regulatory RNA



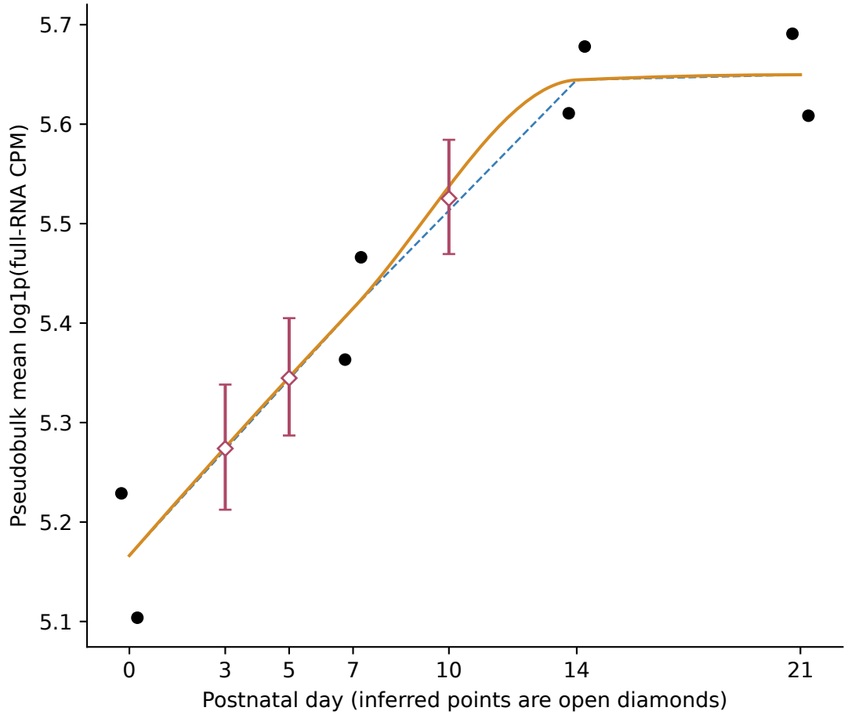
QC EC cycling RNA



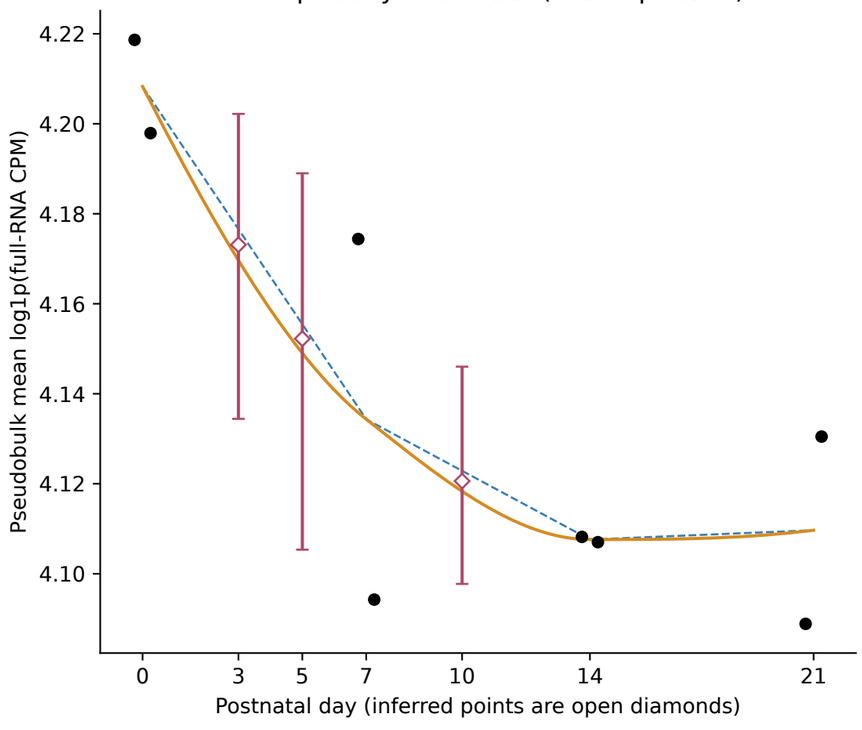
CM cycling RNA



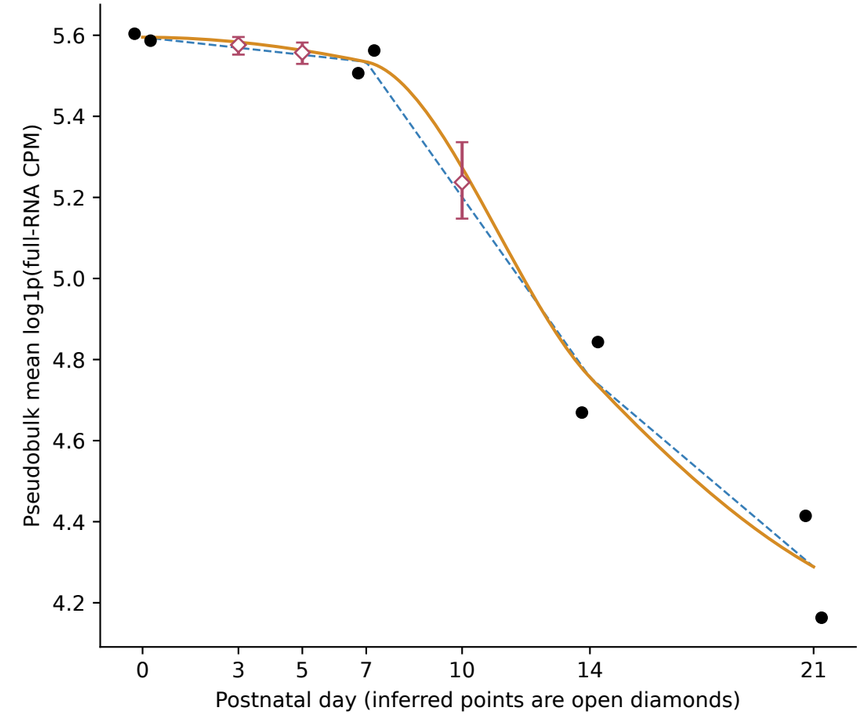
Reassessed FB basement-matrix RNA



CM respiratory-chain RNA (not respiration)



Reassessed FB remodeling RNA



Black points: two observed RNA libraries/age. Open diamonds: model-based expectations, no new samples. Intervals combine endpoint-library bootstrap and two interpolation choices but cannot measure unknown curvature. Respiratory-chain and other RNA panels need not increase together and do not establish function.

Roadmap completed to a testable experiment

Hypothesis selected from these results

The phenotype associated with vascular apposition is developmentally contingent: P7 is a candidate cell-cycle-withdrawal window, whereas P14/P21 show metabolic-regulatory associations. This hypothesis follows component-specific, locally matched observations. It is not a demonstrated causal timing switch, and late low cycling limits detection.

First experiment: oxygen support versus contact

Use a factorial comparison of two measured noninjurious oxygen conditions, direct versus physically separated EC contact, and early versus later initial CM developmental states. Match EC number, matrix, medium, culture duration, baseline CM state and viability. Allocate treatments within independent donor preparations; wells are technical repeats. Pilot at least four donors/state and power confirmation using donor-level variance.

Readouts that decide among mechanisms

Primary: mitochondrial respiration/substrate use and calcium/contractile function. Secondary: sarcomere organization, cycling/withdrawal, ploidy and the corresponding RNA programs. Measure oxygen and assess ROS/DNA damage and survival so injury-associated arrest is not called maturation. An in vivo correlate requires perfused-vessel/hypoxia imaging; culture oxygen is not capillary perfusion.

Decision rules

An oxygen effect without additional contact benefit prioritizes permissive support. A contact benefit at matched oxygen prioritizes an instructive/interface mechanism. Then test EC-specific Ptprm loss/rescue with WT CMs, reciprocal CM perturbation and MEIS1 rescue for a cell-cycle effect. If controlled contact/oxygen effects do not reproduce, reconsider shared regional drivers and other niche cues.

Matrix and remaining reconstruction work

Keep matrix constant first: current apposition-by-FB basement-program interactions are not consistently positive, and RNA is not deposited matrix or stiffness. Exact original Figure 4e replication still requires the unpublished source masks, score-bearing RDS and neighbor table/build settings. Their absence is documented; these derived reconstructions and inferred days do not replace those observations.